Universidad Tecmilenio

2026

MODULE 1

LEARNING OUTCOME 1

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STUDENTS:

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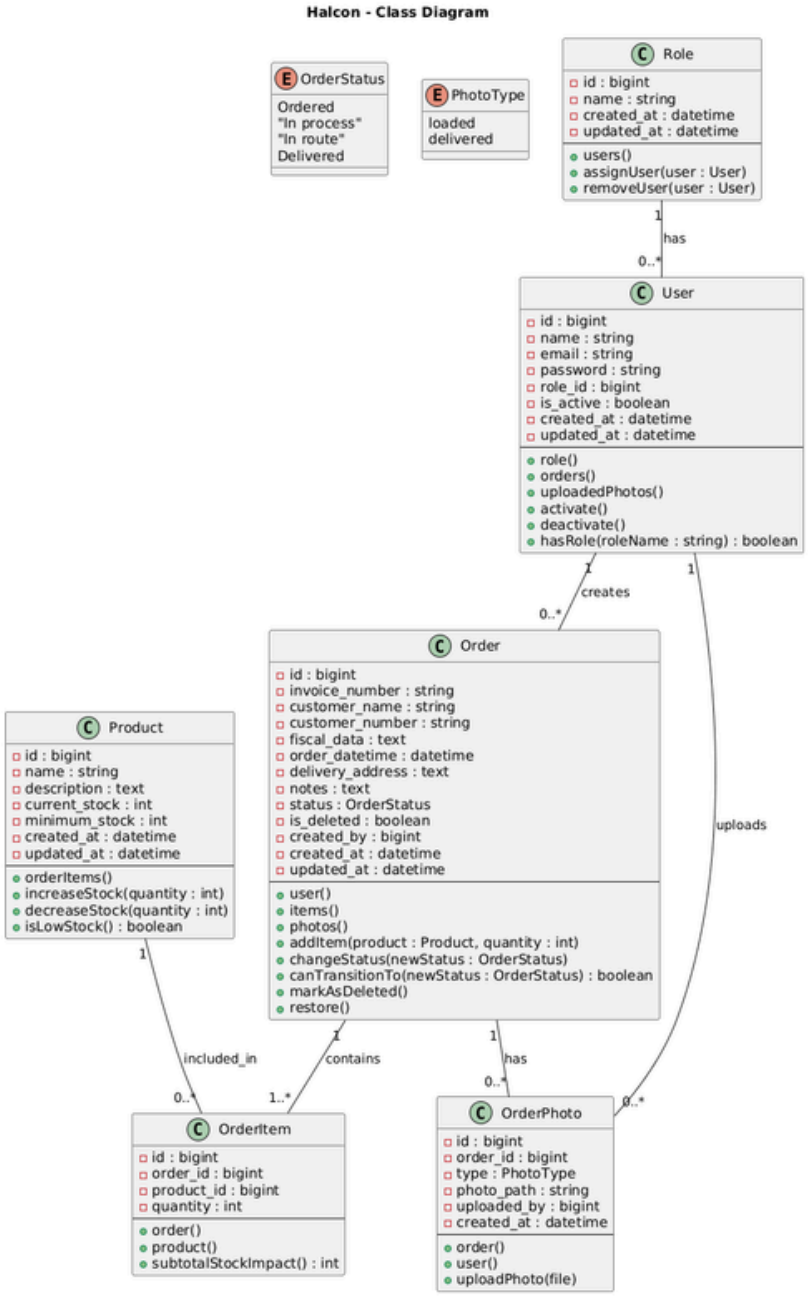
01 Class diagram

1. User & Role Management

- The system includes roles (Admin, Sales, Purchasing, Warehouse, and Route).
- Each user belongs to one role, and the role determines what actions and views are available within the administration panel.

2. Order Management

- The Order class represents customer orders.
- Each order includes invoice number, customer number, delivery address, date, and status.
- Orders follow a mandatory lifecycle:
- Ordered → In process → In route → Delivered.
- An order can contain multiple products through the OrderItem class.



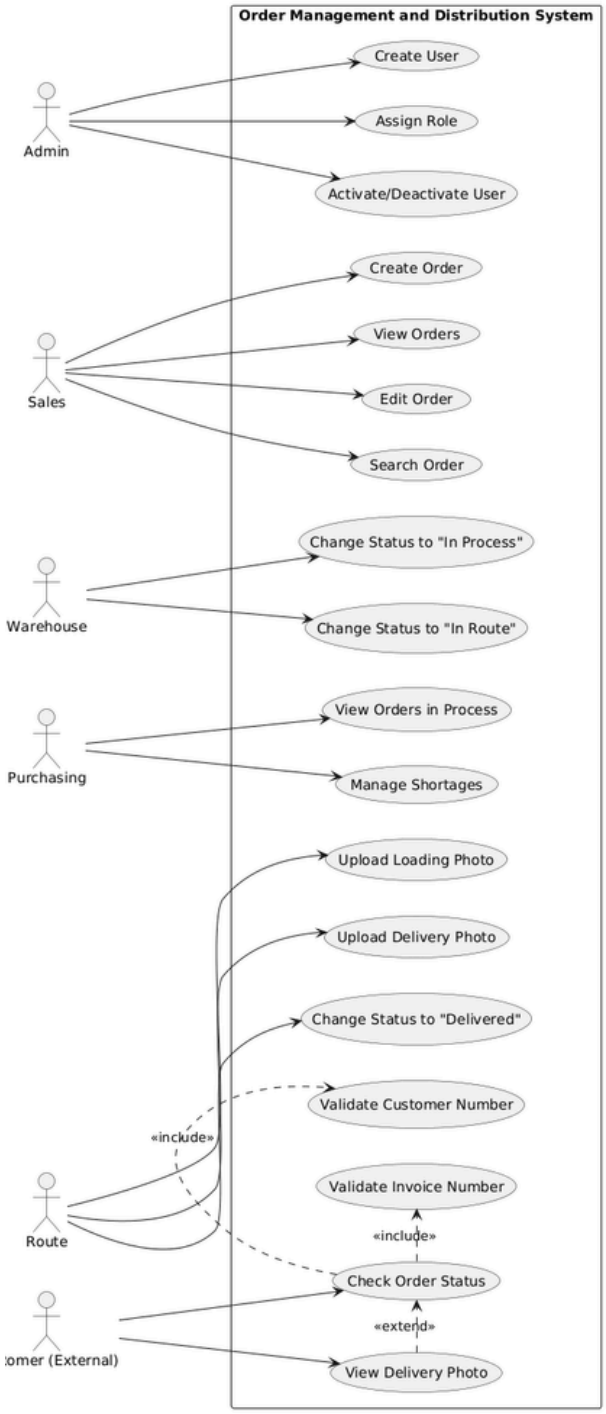
3. Inventory Management

- The Product class manages stock levels.
- Purchasing increases stock, while Warehouse decreases it when preparing orders.

Delivery Evidence

- The OrderPhoto class stores loading and delivery images, uploaded only by the Route role.

02 diagrama



1. User and Role Management

The system implements a role-based access control mechanism to regulate permissions and actions within the platform.

The defined roles are:

- Admin
- Sales
- Purchasing
- Warehouse
- Route

Each user is assigned exactly one role, and the role determines the functionalities available within the system.

The Admin role is responsible for managing system access, including user creation, role assignment, and activation or deactivation of accounts.

This structure ensures controlled access, security, and operational separation of responsibilities.

2. Order Lifecycle Management

The core entity of the system is the Order, which represents a customer request for products and delivery services.

- Each order contains:
- Invoice number
- Customer number
- Delivery address
- Creation date
- Scheduled delivery date
- Current status

Orders follow a mandatory and controlled lifecycle:

Created → In Process → In Route → Delivered

This lifecycle guarantees operational traceability and prevents inconsistent transitions between states.

The responsibilities are distributed as follows:

Sales manages order creation and updates.

Warehouse prepares the order and updates the status to "In Process" and "In Route".

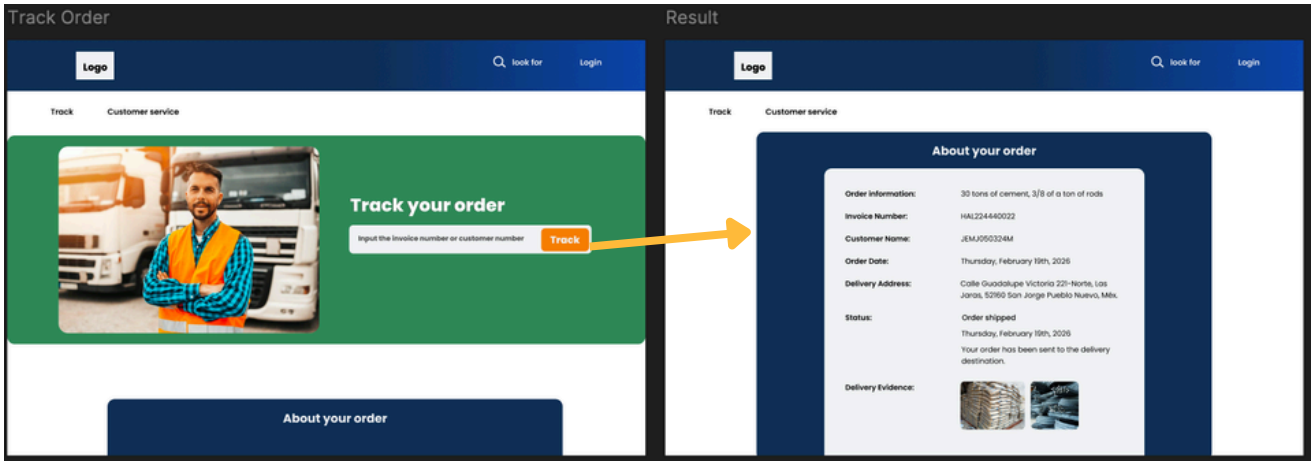
Route finalizes the delivery and updates the status to "Delivered".

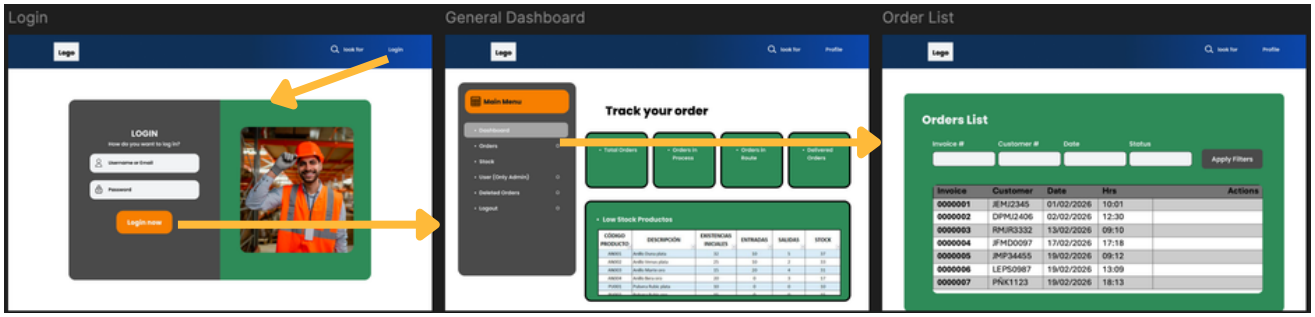
This separation of duties reduces operational errors and improves accountability.

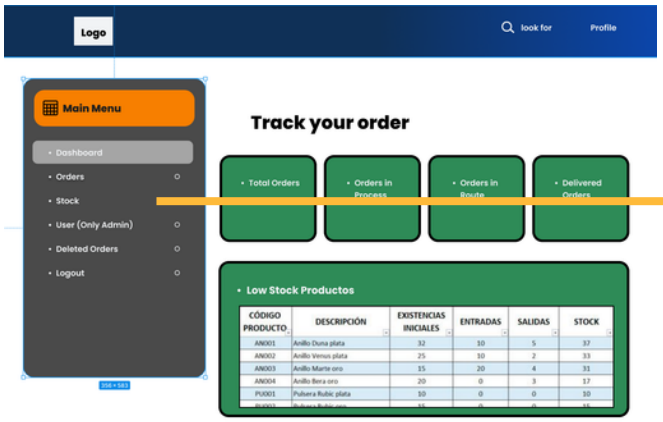
03 Wireframe

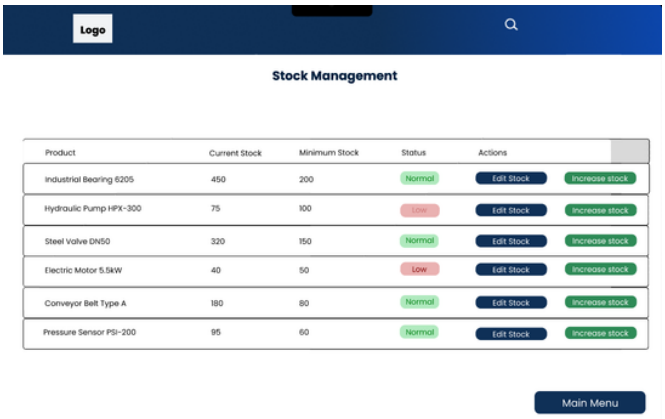
https://www.figma.com/design/zVmPvpwObtT3G7Bg4ZCByP/Untitled?node-id=0-1&t=Y7nLIajJk7DKQHEd-1

1. Screen
2. Track order
3. Login
4. Order Liste






Logo
look for
Profile
Main Menu
Dashboard
Orders
Stock
User (Only Admin)
Deleted Orders
Logout
Track your order
Total Orders
Orders in Process
Orders in Route
Delivered Orders
Low Stock Productos
CÓDIGO PRODUCTO
DESCRIPCIÓN
EXISTENCIAS INICIALES
ENTRADAS
SALIDAS
STOCK
AN001 Anillo Duna plata 32 10 5 37
AN002 Anillo Venus plata 25 10 2 33
AN003 Anillo Marte oro 15 20 4 31
AN004 Anillo Bera oro 20 0 3 17
PU001 Pulsera Rubic plata 10 0 0 10


Logo
Stock Management
Product
Current Stock
Minimum Stock
Status
Actions
Industrial Bearing 6205 450 200 Normal Edit Stock Increase stock
Hydraulic Pump HPX-300 75 100 Low Edit Stock Increase stock
Steel Valve DN50 320 150 Normal Edit Stock Increase stock
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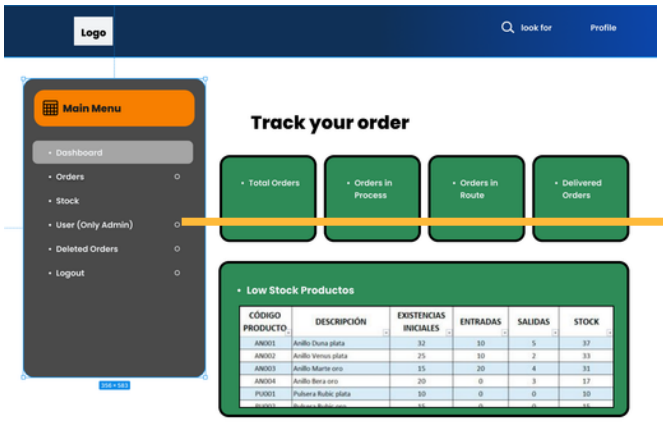
Conveyor Belt Type A 180 80 Normal Edit Stock Increase stock
Pressure Sensor PSI-200 95 60 Normal Edit Stock Increase stock
Main Menu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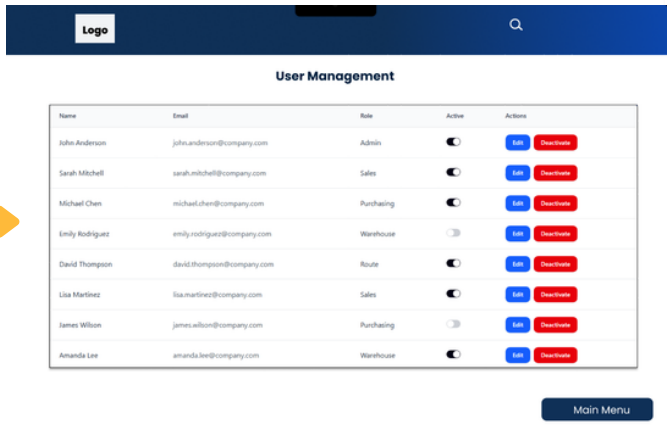
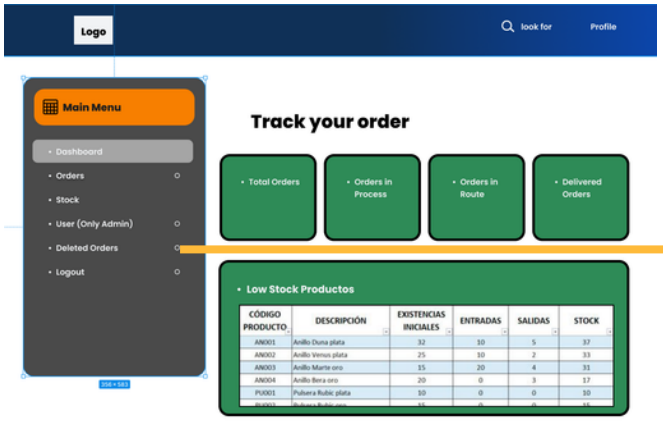

Logo
User Management
Name
Email
Role
Active
Actions
John Anderson john.anderson@company.com Admin Edit Deactivate
Sarah Mitchell sarah.mitchell@company.com Sales Edit Deactivate
Michael Chen michael.chen@company.com Purchasing Edit Deactivate
Emily Rodriguez emily.rodriguez@company.com Warehouse Edit Deactivate
David Thompson david.thompson@company.com Route Edit Deactivate
Lisa Martinez lisa.martinez@company.com Sales Edit Deactivate
James Wilson james.wilson@company.com Purchasing Edit Deactivate
Amanda Lee amanda.lee@company.com Warehouse Edit Deactivate
Main Menu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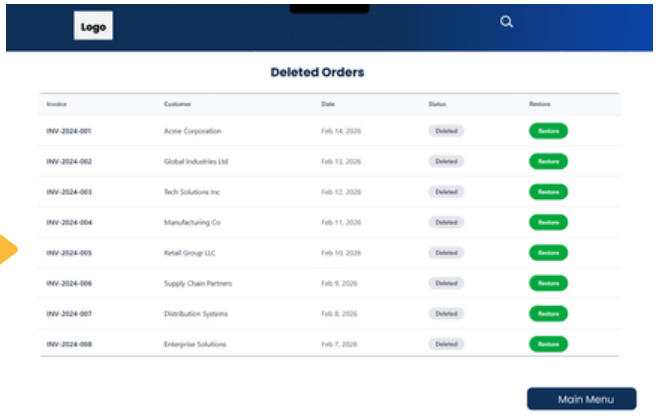
Logo
Deleted Orders
Invoice
Customer
Date
Status
Restore
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INV-2024-008 Enterprise Solutions Feb 7, 2026 Deleted Restore
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